

Physics Olympiad: Computational Challenge

2024: Task 1-9

Tyler Lynch

TASK 1

Task one involves a basic manipulation of the SUVAT equations, to derive an equation for both x and y co-ordinates from a single value of t (time in seconds).

Therefore, this may be considered a parametric model, with t as the parameter.

Where g , ϕ , h , and u are all predetermined inputs, chosen by the user or otherwise 'hard-coded'.

The task description states there is to be no air resistance. Therefore, the only force acting upon the particle is the weight of the particle.

For all current and future equations and derivations, the value of g will remain the positive constant of $9.81ms^{-2}$, unless explicitly stated otherwise.

Both co-ordinates may be derived from the formula:

$$s = ut + \frac{1}{2}at^2$$

Where:

$$u_x = \cos(\phi) u \rightarrow s_x = u_x t$$

$$u_y = \sin(\phi) u \rightarrow s_y = u_y t - \frac{1}{2}gt^2$$

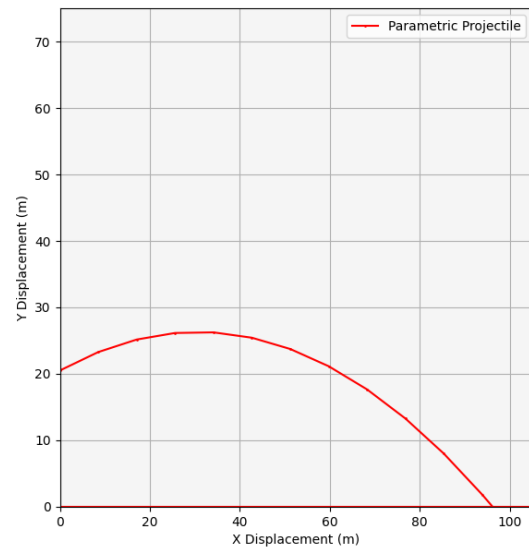
∴ The point p may be equal to:

$$p\left(u_x t, u_y t - \frac{1}{2}gt^2\right)$$

Which is equivalent to:

$$p\left(\cos(\phi) ut, \sin(\phi) ut - \frac{1}{2}gt^2\right)$$

Through this simple derivation, by setting a regular interval of time t (hereon represented as Δt) for an N number of intervals, we can plot N number of x and y points and connect these points with a tangent between each individual point to form a curve, with each x and y co-ordinate forming the vertices of the projectile's parabolic trajectory. The smaller Δt is, the greater the accuracy of the graph.



The graph depicted above depicts a parametric projectile.

Where:

$$u = 30$$

$$\phi = 20$$

$$h = 20$$

$$\Delta t = 0.2$$

TASK 2

Task two requires us to find the apogee of our travelling projectile, as well as a $y(x)$ function, instead of using a parametric model like the previous task. To ensure we have N number of points, we also need to find the maximum x displacement, to then divide by our N value, to get the interval of points.

Deriving the trajectory function

Just like the previous task, we will use the SUVAT equations. h assumed to be 0:

$$s = ut + \frac{1}{2}at^2$$

Where:

$$u_x = \cos(\theta) u \rightarrow s_x = u_x t$$

$$u_y = \sin(\theta) u \rightarrow s_y = u_y t - \frac{1}{2}gt^2$$

Let:

$$u_x = x, \quad u_y = y$$

Now, we rearrange and substitute.

$$\frac{x}{u_x} = t$$

$$y = u_y \left(\frac{x}{u_x} \right) - \frac{1}{2}g \left(\frac{x}{u_x} \right)^2$$

Substitute:

$$u_x = \cos(\theta) u$$

$$u_y = \sin(\theta) u$$

$$y = \sin(\theta) u \left(\frac{x}{\cos(\theta) u} \right) - \frac{1}{2}g \left(\frac{x}{\cos(\theta) u} \right)^2$$

$$y = \left(\frac{\sin(\theta) u}{\cos(\theta) u} \right) x - \left(\frac{g}{2\cos(\theta)^2 u^2} \right) x^2$$

$$y = \tan(\theta) x - \left(\frac{g}{2\cos(\theta)^2 u^2} \right) x^2$$

Therefore:

$$f(x) = \tan(\theta) x - \left(\frac{g}{2\cos(\theta)^2 u^2} \right) x^2$$

In the condition where $h \neq 0$ and is real:

$$f(x) = \tan(\theta) x - \left(\frac{g}{2\cos(\theta)^2 u^2} \right) x^2 + h$$

Finding the maximum range of a particle

To find the maximum range of a particle with a given angle, velocity, and height, we can once again use SUVAT.

$$x_{max} = u_x t_{max}$$

To solve for s_x find t_{max} :

$$y = \frac{-1}{2}gt_{max}^2 + u_y t_{max} + h$$

Find x -axis intersection, let $y = 0$:

$$0 = \frac{1}{2}gt_{max}^2 - u_y t_{max} - h$$

$$t_{max} = \frac{u_y + \sqrt{u_y^2 + 2gh}}{g}$$

Therefore:

$$x_{max} = u_x \frac{u_y + \sqrt{u_y^2 + 2gh}}{g}$$

$$x = \frac{\sin(\theta) \cos(\theta) u^2 + \sqrt{\sin^2(\theta) u^2 + 2gh}}{g}$$

Finding the apogee's x co-ordinate

To find the apogee, we are finding the vertex of the parabola. We may either complete the square or differentiate. Differentiating proves to be simpler however, as it removes the h constant.

Using our derived function:

$$y = \tan(\theta) x - \left(\frac{g}{2\cos(\theta)^2 u^2} \right) x^2 + h$$

$$\frac{dy}{dx} = \tan(\theta) - \left(\frac{g}{\cos(\theta)^2 u^2} \right) x$$

Assume $\frac{dy}{dx} = 0$:

$$\tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)} = \left(\frac{g}{\cos(\theta)^2 u^2} \right) x$$

$$x = \frac{\sin(\theta) \cos(\theta) u^2}{g}$$

$$\frac{\sin(\emptyset) \cos(\emptyset) u^2}{g} \equiv \frac{\sin(2\emptyset) u^2}{2g}$$

Apogee Substitution

$$\begin{aligned} f(x) &= \tan(\emptyset) \left(\frac{\sin(\emptyset) \cos(\emptyset) u^2}{g} \right) \\ &\quad - \left(\frac{g}{2\cos(\emptyset)^2 u^2} \right) \left(\frac{\sin(\emptyset) \cos(\emptyset) u^2}{g} \right)^2 + h \\ f(x) &= \left(\frac{2\sin^2(\emptyset) u^2}{2g} \right) - \left(\frac{\sin^2(\emptyset) u^2}{2g} \right) + h \\ f(x) &= \left(\frac{2\sin^2(\emptyset) u^2 - \sin^2(\emptyset) u^2}{2g} \right) + h \\ f(x) &= \left(\frac{\sin^2(\emptyset) u^2}{2g} \right) + h \end{aligned}$$

The apogee, A, is therefore equal to:

$$A \left(\frac{\sin(2\emptyset) u^2}{2g}, \frac{\sin^2(\emptyset) u^2}{2g} + h \right)$$

Finding the value of the x-points

To plot a graph of the analytic model, we need an array of x points. Assume there to be N number of x points

Using our max range formula:

$$x = \frac{\sin(\emptyset) \cos(\emptyset) u^2 + \sqrt{\sin^2(\emptyset) u^2 + 2gh}}{g}$$

Where $x_{max} = x$:

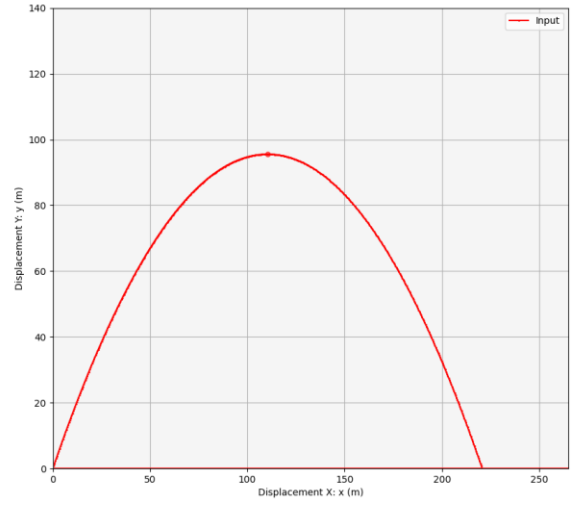
$$\Delta x = \frac{x_{max}}{N}$$

$$x_{n+1} = x_n + \Delta x$$

For all N number of x points:

$$y_n = f(x_n) = \tan(\emptyset) x - \left(\frac{g}{2\cos(\emptyset)^2 u^2} \right) x^2$$

Plot Application



Graphical plot of the analytic model, with parameter:

$$u = 50$$

$$h = 0$$

$$\emptyset = 60$$

With apogee of:

$$A(110,96)$$

With x_{max} of:

$$x_{max} = 221$$

TASK 3

Task three requires us find the maximum, and minimum angles of a particle going through a predetermined point P . It also requires us to find the graph of minimum velocity.

Let x_p and y_p represent the respective x and y co-ordinates of point P relative to the launch height h (so that $y_p = y - h$).

Highball & lowball angles

$$y_p = \tan(\theta) x - \left(\frac{g}{2\cos(\theta)^2 u^2} \right) x_p^2$$

Use trigonometric identity:

$$y_p = \tan(\theta) x_p - \left(\frac{gx_p^2}{2u^2} \right) (1 + \tan^2(\theta))$$

$$= \tan(\theta) x_p - \left(\frac{gx_p^2}{2u^2} + \frac{gx_p^2}{2u^2} \tan^2(\theta) \right)$$

$$y_p = \tan(\theta) x_p - \frac{gx_p^2}{2u^2} - \frac{gx_p^2}{2u^2} \tan^2(\theta)$$

$$0 = \frac{gx_p^2}{2u^2} \tan^2(\theta) - \tan(\theta) x_p + \left[\frac{gx_p^2}{2u^2} + y_p \right]$$

Solve for $\tan(\theta)$. Let $\frac{gx_p^2}{u^2} = M$:

$$\tan(\theta) = \frac{x_p \pm \sqrt{x_p^2 - 2M \left(\frac{1}{2}M + y_p \right)}}{M}$$

Therefore:

$$\theta = \tan^{-1} \left(\frac{x_p \pm \sqrt{x_p^2 - 2M \left(\frac{M}{2} + y_p \right)}}{M} \right)$$

Minimum velocity

$$0 = \frac{gx_p^2}{2u^2} \tan^2(\theta) - \tan(\theta) x_p + \left[\frac{gx_p^2}{2u^2} + y_p \right]$$

There may only be one root where $u = u_{min}$, therefore find the discriminant:

$$0 = x^2 - \frac{gx_p^2}{u_{min}^2} \left(\frac{gx_p^2}{u_{min}^2} + 2y_p \right)$$

$$0 = x^2 - \frac{g^2 x_p^4}{u_{min}^4} - \frac{2gx_p^2 y_p}{u_{min}^2}$$

$$0 = u_{min}^4 x^2 - g^2 x_p^4 - 2u_{min}^2 g y_p x_p^2$$

$\because x \neq 0$:

$$\therefore 0 = u_{min}^4 - 2u_{min}^2 g y_p - g^2 x_p^2$$

$$0 = u_{min}^4 - 2u_{min}^2 g y_p - g^2 x_p^2$$

Solve for u , complete the square:

$$g^2 y_p^2 + g^2 x_p^2 = (u_{min}^2 - g y_p)^2$$

$$g^2 (y_p^2 + x_p^2) = (u_{min}^2 - g y_p)^2$$

$$g \sqrt{y_p^2 + x_p^2} = u_{min}^2 - g y_p$$

$$\sqrt{g} \sqrt{y_p + \sqrt{y_p^2 + x_p^2}} = u_{min}$$

Minimum angle

Use the previous formula to find highball and lowball angles, substitute $u = u_{min}$

$$\sqrt{g} \sqrt{y_p + \sqrt{y_p^2 + x_p^2}} = u_{min}$$

Let $\frac{gx_p^2}{u_{min}^2} = M_{min}$:

$$\tan(\theta) = \frac{x_p + \sqrt{x_p^2 - 2M_{min} \left(\frac{M_{min}}{2} + y_p \right)}}{M_{min}}$$

TASK 4

Task 4 requires us to find the angle ϕ_R at which a particle of velocity u travels the maximum possible range R , with a given velocity u .

Therefore $R = x$ when $y = 0$

Finding ϕ_R

$$0 = \tan(\phi) R - \left(\frac{\sec^2(\phi) g}{2u^2} \right) R^2 + h$$

Finding the maximum of R , therefore differentiate.

$$\text{Let } L = \frac{u^2}{g}:$$

$$\frac{dy}{dR} = R \sec^2(\phi) - \frac{R^2 \sec^2(\phi) \tan(\phi)}{L}$$

$$\frac{dy}{dR} = R \sec^2(\phi) - \frac{R^2 \sec^2(\phi) \tan(\phi)}{L}$$

$$\frac{dy}{dR} = R \sec^2(\phi) - \frac{gR^2 \sec^2(\phi) \tan(\phi)}{u^2} = 0$$

$$R \sec^2(\phi) = \frac{gR^2 \sec^2(\phi) \tan(\phi)}{u^2}$$

$$1 = \frac{gR \tan(\phi)}{u^2}$$

$$\frac{u^2}{gR} = \tan(\phi)$$

Finding R

Using the following version of the projectile trajectory function:

$$\tan(\phi) R - \left(\frac{gR^2}{2u^2} \right) (1 + \tan^2(\phi)) + h$$

Substitute $\tan(\phi)$:

$$\frac{u^2}{gR} R - \left(\frac{gR^2}{2u^2} \right) \left(1 + \frac{u^4}{g^2 R^2} \right) + h = 0$$

$$\frac{u^2}{g} - \left(\frac{gR^2}{2u^2} \right) \left(1 + \frac{u^4}{g^2 R^2} \right) + h =$$

$$\frac{u^2}{g} - \frac{gR^2}{2u^2} - \frac{u^4}{2gu^2} + h =$$

$$\frac{u^2}{g} + h = \frac{gR^2}{2u^2} + \frac{u^4}{2gu^2}$$

$$\frac{u^2}{g} + h = \frac{gR^2}{2u^2} + \frac{u^4}{2gu^2}$$

$$2 \left(\frac{u^2}{g} + h \right) = \frac{gR^2}{u^2} + \frac{u^4}{gu^2}$$

$$2gu^2 \left(\frac{u^2}{g} + h \right) = g^2 R^2 + u^4$$

$$\frac{u^2 \left(2g \left(\frac{u^2}{g} + h \right) - u^2 \right)}{g^2} = R^2$$

$$\frac{u \sqrt{2g \left(\frac{u^2}{g} + h \right) - u^2}}{g} = R$$

$$\frac{u}{g} \sqrt{2u^2 - u^2 + 2gh} = R$$

Therefore:

$$\frac{u}{g} \sqrt{u^2 + 2gh} = R$$

TASK 5

Task 5 requires us to find a graph which contains all the possible trajectories with a given value of u

Solve the discriminant of the following equation, for y :

$$\text{Let } \frac{gx^2}{u^2} = M$$

$$\tan(\phi) = \frac{x \pm \sqrt{x^2 - 2M \left(\frac{1}{2} M + y \right)}}{M}$$

$$1 - \frac{2g}{u^2} \left(\frac{gx^2}{2u^2} + y \right) = 0$$

$$1 - \frac{g^2 x^2}{u^4} = y \frac{2g}{u^2}$$

$$\frac{u^2}{2g} - \frac{x^2}{2u^2} = y$$

Therefore, the previous equation describes the bounding parabola, also known as safety parabola or the envelope of trajectories.

BPhO: Computational Challenge

TASK 6

Task 6 involves finding the distance d travelled by a particle, with given velocity, angle, and height.

This can be approximated by using the following equation:

$$\sum \sqrt{\Delta x^2 + \Delta y^2} = d$$

As the distance between subsequent values approaches 0:

$$\lim_{\Delta \rightarrow 0} \sqrt{\Delta x^2 + \Delta y^2} = \int \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

Finding $\frac{dy}{dx}$

$$y = \tan(\theta) x - \left(\frac{g}{2u^2 \cos^2(\theta)}\right) x^2 + h$$

$$\frac{dy}{dx} = \tan(\theta) - \frac{gx}{u^2} (1 + \tan^2(\theta))$$

Substituting $\frac{dy}{dx}$

$$\int_0^{x_{max}} \sqrt{1 + \left(\tan(\theta) - \frac{gx}{u^2} (1 + \tan^2(\theta))\right)^2} dx$$

Let $\alpha = \tan(\theta) - \frac{gx}{u^2} (1 + \tan^2(\theta))$:

$$\frac{-u^2}{g(1 + \tan^2(\theta))} \int_{\tan(\theta)}^{\alpha_{max}} \sqrt{1 + \alpha^2} = d$$

Apply standard integral:

$$\int \sqrt{1 + x^2} = \frac{1}{2} \ln(\sqrt{1 + x^2} + x) + \frac{1}{2} x \sqrt{1 + x^2} + c$$

To obtain:

$$\frac{-u^2}{g(1 + \tan^2(\theta))} \left[\frac{1}{2} \ln(\sqrt{1 + x^2} + x) + \frac{1}{2} x \sqrt{1 + x^2} \right]_{\tan(\theta)}^{\alpha_{max}} = d$$

Remove the negative, therefore flip the bounds:

$$\frac{u^2}{g(1 + \tan^2(\theta))} \left[\frac{1}{2} \ln(\sqrt{1 + x^2} + x) + \frac{1}{2} x \sqrt{1 + x^2} \right]_{\alpha_{max}}^{\tan(\theta)} = d$$

TASK 7

Task 7 requires us to measure the distance d of a moving particle from the origin, with respect to time t .

The launch height h is assumed to be 0, as it is relative to the launch point, therefore it has no bearing on the result.

$$\sqrt{y^2 + x^2} = d$$

$$x = u \cos(\theta) t$$

$$y = u \sin(\theta) t - \frac{1}{2} g t^2$$

$$\sqrt{\left(u \sin(\theta) t - \frac{1}{2} g t^2\right)^2 + (u \cos(\theta) t)^2} = d$$

$$\sqrt{u^2 \sin^2(\theta) t^2 - u \sin(\theta) g t^3 + \frac{1}{4} g^2 t^4 + u^2 \cos^2(\theta) t^2}$$

$$\sqrt{u^2 t^2 (\sin^2 \theta + \cos^2 \theta) - g t^3 u \sin(\theta) + \frac{1}{4} g^2 t^4}$$

$$\sqrt{u^2 t^2 - g t^3 u \sin(\theta) + \frac{1}{4} g^2 t^4} = d$$

$$t \sqrt{u^2 - g t u \sin(\theta) + \frac{1}{4} g^2 t^2} = d$$

Finding the minima and maxima

$$u^2 t^2 - g t^3 u \sin(\theta) + \frac{1}{4} g^2 t^4 = d^2$$

$$\frac{d(d^2)}{dt} = 2u^2 t - 3g t^2 u \sin(\theta) + g^2 t^3$$

$$0 = t(2u^2 - 3g t u \sin(\theta) + g^2 t^2)$$

Where $t > 0$:

$$0 = 2u^2 - 3g t u \sin(\theta) + g^2 t^2$$

BPhO: Computational Challenge

Complete the square:

$$0 = 2u^2 - 3gtu \sin(\phi) + g^2 t^2$$

$$0 = t^2 - \frac{3tu}{g} \sin(\phi) + \frac{2u^2}{g^2}$$

$$\frac{9u^2}{4g^2} \sin^2(\phi) - \frac{2u^2}{g^2} = \left(t - \frac{3u}{2g} \sin(\phi)\right)^2$$

$$\frac{9u^2 \sin^2(\phi)}{4g^2} - \frac{8u^2}{4g^2} = \left(t - \frac{3u}{2g} \sin(\phi)\right)^2$$

$$\pm \sqrt{\frac{9u^2 \sin^2(\phi) - 8u^2}{4g^2}} = t - \frac{3u}{2g} \sin(\phi)$$

$$\frac{3u}{2g} \sin(\phi) \pm \frac{u}{2g} \sqrt{9 \sin^2(\phi) - 8} = t$$

This gives us the t value of the minima/maxima point.

Substituting t

Letting

$$\frac{3u}{2g} \sin(\phi) \pm \frac{u}{2g} \sqrt{9 \sin^2(\phi) - 8} = t:$$

And then substituting into our distance formula gives us the d value of the turning point.

$$t \sqrt{u^2 - gtu \sin(\phi) + \frac{1}{4} g^2 t^2} = d$$

Therefore, giving us the distance at which the turning point occurs.

TASK 8

Task 8 involves the use of Verlet Integration. Verlet integration, despite the name, isn't integration, but rather a numerical method used to achieve results akin to integrating Newton's equations of motion. It's commonly used for air resistance models, but that appears in **task 9**. Instead, we need to create a projectile that bounces when it collides with the x-axis. This task, mathematically, is much easier, as essentially, we are just substituting numbers into a formula.

The Verlet Method

Firstly, we need to establish our projectile in space, we do this with the positional vector p

$$p = \begin{pmatrix} 0 \\ h \end{pmatrix}$$

Where h is the launch height.

We then need to translate our launch angle ϕ and our launch velocity v_L into another 2-dimensional vector.

$$v = \begin{pmatrix} v_L \cos(\phi) \\ v_L \sin(\phi) \end{pmatrix}$$

And then our acceleration vector:

$$a = \begin{pmatrix} 0 \\ -g \end{pmatrix}$$

Now, assuming there to be a constant different of t , equal to Δt , we create a function to return the position, and the velocity

$$p_{n+1} = p_n + v_n \Delta t + \frac{1}{2} a_n (\Delta t)^2$$

$$v_{n+1} = v_n + a \Delta t$$

Mathematically, this is it. These two functions are iterated until the conditions are satisfied. The conditions, also, are easy to program.

The conditions, explained logically, are:

If $p_x \leq 0$ and $b < n$:

$$b = b + 1$$

$$v_y = -v_y c$$

Where b is the number of bounces, n is the max number of bounces, and c is the coefficient or restitution. c is less than 1, and it reduces the velocity of a projectile upon impact, mimicking the transfer of energy from a projectile to its surface in a real-world scenario.

TASK 9

Task 9 is the final task, with the exclusion of extensions. It is very similar to task 8, but instead of a bouncing projectile, it requires the inclusion of air resistance.

This therefore means that additional forces, besides weight, will be acting upon the projectile. We will need to therefore change the acceleration vector every iteration.

Resistive Verlet Method

Initially, we need to create a function that alters the acceleration according to the velocity.

Determining our air resistance factor K :

$$K = \rho C A$$

Where ρ is the fluid density (fluid density of air changes with altitude, I am using 1.225), C is the drag coefficient, and A is the cross-sectional area.

We then create a formula for the drag force F :

$$F = \frac{1}{2} K v^2$$

Therefore, the drag acceleration is equal to:

$$a = -\frac{K v^2}{2m}$$

However, this is only in the absence of gravity. In reality, it is instead:

$$\begin{pmatrix} a_x \\ a_y \end{pmatrix} = \begin{pmatrix} -\frac{K v_x^2}{2m} \\ -gm - \frac{K v_y^2}{2m} \end{pmatrix}$$

Now, we have our 3 functions, for position, velocity, and acceleration. We can simply iterate them in this sequence.

Iteration n :

$$p_{n+1} = p_n + v_n \Delta t + \frac{1}{2} a_n (\Delta t)^2$$

$$\begin{pmatrix} a_x \\ a_y \end{pmatrix} = \begin{pmatrix} -\frac{K v_x^2}{2m} \\ -gm - \frac{K v_y^2}{2m} \end{pmatrix}$$

$$v_{n+1} = v_n + a \Delta t$$

And plot the position x and position y to create our displacement graph.